Result 1 (endpoint constructor). [Topology foundations, endpoint constructor]
Let a and b be real numbers. Let $I = \text{endpoints}(a, b)$. Then

$$I.\text{lower} = a \text{ and } I.\text{upper} = b.$$

Result 2 (shifting endpoints preserves order). [Topology foundations, translating ordered endpoints]
Let I be a pair of real endpoints. Let s be a real number. Assume ordered endpoints(I) (`ordering of I`). Then

$$\text{ordered endpoints}(\text{shifted endpoints}(I, s)).$$

Result 3 (bundled path has its declared endpoints). [Topology foundations, dependent path projection]
Let X be a type. Let P be a path with its topology and endpoints on X. Then

$$\left(\text{there exists } p : \text{Path}_{P.\text{topology}}(P.\text{start}, P.\text{finish}),\ p = P.\text{path}\right).$$

Result 4 (reversing a bundled path twice). [Topology foundations, bundled path involution]
Let X be a type. Let P be a path with its topology and endpoints on X. Then

$$\text{reversed bundle}(X, \text{reversed bundle}(X, P)) = P.$$

Result 5 (function field application). [Topology foundations, function field application]
Let f be a function from $\mathbb{R}$ to $\mathbb{R}$. Let x be a real number. Let $P = \text{scalar map}(f)$. Then

$$P.\text{map}(x) = f(x).$$

Result 6 (vector field dimensions). [Topology foundations, natural field dependency]
Let V be an element of SizedVector. Then

$$\left(\text{there exists } v : \mathbb{R}^{V.\text{size}},\ v = V.\text{vector}\right).$$

Result 7 (computed constructor projection). [Topology foundations, constructor field computation]
Let a and b be real numbers. Then

$$\text{endpoints}(a, b).\text{lower} = a.$$

Result 8 (authored record projection). [Topology foundations, translated endpoints]
Let I be a pair of real endpoints. Let s be a real number. Then

$$\text{shifted endpoints}(I, s).\text{lower} = I.\text{lower} + s.$$

Result 9 (opaque record projection). [Topology foundations, native identity adapter]
Let I be a pair of real endpoints. Then

$$I.\text{lower} = I.\text{lower}.$$

Result 10 (computed dependent field). [Topology foundations, dependent constructor projection]
Let X be a type. Let P be a path with its topology and endpoints on X. Then

$$\text{based path}(X, P.\text{topology}, P.\text{start}, P.\text{finish}, P.\text{path}).\text{path} = P.\text{path}.$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.